

ros2 bag 로 검증 파이프라인 구성

ros2 bag record를 통해 받은 메시지를 .mcap 파일에 순서대로 저장하여 기록한다.

ros2 bag play는 .mcap 파일을 읽으면서 기록 당시 저장해둔 타임스탬프 간격으로 메시지를 보낸다.

```
yepunman@yepunman: ~  
yepunman@yepunman:~$ ros2 topic echo /topic  
WARNING: topic [/topic] does not appear to be published yet  
Could not determine the type for the passed topic  
yepunman@yepunman:~$ ros2 topic echo /topic std_msgs/msg/String  
data: Hello World!  
---  
data: Hello World!  
---  
data: Hello World!  
---  
data: Hello World!  
---  
data: Hello World!  
---  
yepunman@yepunman:~$ ros2 bag record --output ~/ros2_ws/bagdemo  
[INFO] [1789460228.424473174] [rosbag2_recorder]: Event publisher thread: Started  
[INFO] [1789460228.425635793] [rosbag2_recorder]: Starting recording to '/home/yepunman/ros2_ws/bagdemo'  
[INFO] [1789460379.617944858] [rosbag2_recorder]: Listening for topics...  
[INFO] [1789460379.621764227] [rosbag2_recorder]: Recording...  
[INFO] [1789460379.621794152] [rosbag2_recorder]: Topics discovery started.  
[INFO] [1789460379.621801765] [rosbag2_recorder]: Subscribed to topic '/topic'  
[INFO] [1789460379.621808139] [rosbag2_recorder]: All requested topics are subscribed. Stopping discovery...  
[INFO] [1789460379.621814495] [rosbag2_recorder]: Topics discovery stopped.  
[INFO] [1789460379.623004584] [rosbag2_recorder]: Pausing recording.  
[INFO] [1789460379.623004584] [rosbag2_recorder]: Writing remaining messages from cache to the bag. It may take a while  
[INFO] [1789460379.623004584] [rosbag2_recorder]: Recording stopped  
[INFO] [1789460379.623004584] [rosbag2_recorder]: Event publisher thread: Exited  
yepunman@yepunman:~$ ros2 bag play ~/ros2_ws/bagdemo  
[INFO] [1789460461.776981315] [rosbag2_player]: Set rate to 1  
[INFO] [1789460461.780292654] [rosbag2_player]: Adding keyboard callbacks.  
[INFO] [1789460461.780319841] [rosbag2_player]: Press SPACE for Pause/Resume  
[INFO] [1789460461.780326626] [rosbag2_player]: Press CURSOR_RIGHT for Play Next Message  
[INFO] [1789460461.780341407] [rosbag2_player]: Press CURSOR_UP for Increase Rate 10%  
[INFO] [1789460461.780348310] [rosbag2_player]: Press CURSOR_DOWN for Decrease Rate 10%  
[INFO] [1789460461.781526755] [rosbag2_player]: Playback until timestamp: -1  
yepunman@yepunman:~$
```

무정수 값을 담아 시작연산이 가능한 계산기 서비스로 확장
라미터로 도자 선저 인브하

```
yepunman@yepunman: ~  
yepunman@yepunman:~$ ros2 bag info ~/ros2_ws/bagdemo  
Files: bagdemo_0.mcap  
Bag size: 6.2 KiB  
Storage id: mcap  
ROS Distro: Jazzy  
Duration: 0.500153155s  
Start: Sep 15 2026 17:42:52.158717023 (1789461772.158717023)  
End: Sep 15 2026 17:43:00.658870178 (1789461780.658870178)  
Messages: 18  
Topic information: Topic: /topic | Type: std_msgs/msg/String | Count: 18 | Serialization format: cdr  
Service: 0  
Service information:  
yepunman@yepunman:~$  
yepunman@yepunman:~$ ros2 topic echo /topic std_msgs/msg/String  
data: Hello World!  
---  
data: Hello World!  
---  
data: Hello World!  
---  
data: Hello World!  
---  
data: Hello World!  
---  
data: Hello World!  
---  
yepunman@yepunman:~$  
yepunman@yepunman:~$ source /opt/ros/jazzy/setup.bash && source ~/ros2_ws/install/setup.bash  
yepunman@yepunman:~$ rm -rf ~/ros2_ws/bagdemo  
yepunman@yepunman:~$ ros2 bag play ~/ros2_ws/bagdemo  
[INFO] [1789461839.092398658] [rosbag2_player]: Set rate to 1  
[INFO] [1789461839.095554694] [rosbag2_player]: Adding keyboard callbacks.  
[INFO] [1789461839.095581829] [rosbag2_player]: Press SPACE for Pause/Resume  
[INFO] [1789461839.095587552] [rosbag2_player]: Press CURSOR_RIGHT for Play Next Message  
[INFO] [1789461839.095592065] [rosbag2_player]: Press CURSOR_UP for Increase Rate 10%  
[INFO] [1789461839.095596643] [rosbag2_player]: Press CURSOR_DOWN for Decrease Rate 10%  
[INFO] [1789461839.096808519] [rosbag2_player]: Playback until timestamp: -1  
yepunman@yepunman:~$
```

ros2 topic echo를 하면 기록한 메시지를 재생할 수 있는데 처음 사진왼쪽 터미널에 보면 Hello world!가 출력되는 것을 볼수 있다.

두번째 사진을 보면 ros2 bag info를 사용하면 messege: N을 통해 몇개의 메시지를 입력받았는지 확인할 수 있다.

ros2 bag play를 하면 스페이스바 일시정지, 방향키로 스텝/속도조절와 같은 기능을 키보드로 사용할 수 있다.